

Preface

Hi there – I am Daniel, an entrepreneur, technical advisor, and full-stack developer with a focus on technologies in the React ecosystem.

In my time as a technical advisor and developer for enterprises and the public sector, I have noticed that more and more companies look to reduce the gap between frontend and backend developers. Their business requirements often result in the need for a so-called “backend for frontend,” where data is fetched from different backend systems and prepared in a way that can be easily displayed in the frontend.

As an entrepreneur, I also have experience with starting new projects with smaller teams, where it is essential that every developer on your team can do everything, not just the frontend or the backend. In such cases, it often makes sense to develop the backend and frontend in the same language, which is often JavaScript (or TypeScript), because there is a big ecosystem and a large number of developers available.

In both cases, becoming a full-stack developer is getting increasingly more important. I have been coaching developers to learn more about full-stack development for a long time and have noticed that there are common issues and misunderstandings that most developers encounter when learning full-stack development. In this book, I want to summarize all my learnings and teachings about full-stack development, giving you pointers on where and how to learn more about the ever-growing ecosystem of full-stack development in JavaScript.

Nowadays, many companies use a stack consisting of MongoDB, Express, React, and Node.js, called the MERN stack. In this book, I will teach you how to build modern full-stack React applications by using these technologies. I will teach these technologies from the ground up, using as few

libraries as possible, so that you can learn the essential concepts. This will allow you to adapt to new technologies for years to come, even when the specific tools used in this book become outdated. Additionally, I will teach about the deployment of apps and DevOps, as I found that this sector is often neglected and there are not enough developers who know about it. In the last part of the book, I will introduce Next.js as a full-stack framework and give an outlook on new developments in this sector, such as React Server Components and Server Actions.

I hope you enjoy reading this book. If you have any questions or feedback, feel free to reach out to me!

Who this book is for

This book is for developers who already have experience with React and want to learn how to create, integrate, and deploy various backend systems to become a full-stack developer. You should already have a good understanding of JavaScript and React, but do not need to have any prior knowledge of the development, creation, integration, and deployment of backend systems. If you face one of the following challenges, this book will be perfect for you:

- You know how to make a frontend with React but have no idea how to properly integrate it with a backend
- You want to create a full-stack project from scratch but do not know how to
- You want to learn more about the deployment of apps and DevOps
- You want to learn more about modern React development, such as React Server Components, Server Actions, and Next.js

This book will provide you with real-world projects and includes all the steps needed to become a full-stack developer, including but not limited to backend development, frontend development, testing (unit tests and end-to-end tests), and deployment.

What this book covers

Chapter 1, *Preparing for Full-stack Development*, gives a brief overview of the contents of the book and teaches you how to set up a project that will be used as a basis for the development of your full-stack projects.

Chapter 2, *Getting to Know Node.js and MongoDB*, provides information on how to write and run scripts with Node.js. Then, it explains how to use Docker to set up a database service. It also introduces MongoDB, a document database, and how to access the MongoDB database via Node.js.

Chapter 3, *Implementing a Backend Using Express*, puts into practice what you learned in **Chapter 2** by creating a backend service. Express is used to provide a REST API, Mongoose ODM to interface with MongoDB, and Jest to write unit tests for the backend code.

Chapter 4, *Integrating a Frontend Using React and TanStack Query*, provides instructions on how to create a frontend that interfaces with the previously created backend service. It uses Vite to set up a React project, in which we create a basic user interface. Then, it teaches you how to use TanStack Query, a data-fetching library, to handle backend state and integrate the backend API with the frontend.

Chapter 5, *Deploying the Application with Docker and CI/CD*, deep-dives into DevOps by teaching you about Docker and how to package an app with it. Then, it provides instructions on how to deploy an app to a cloud provider and how to configure CI/CD to automate the deployment.

Chapter 6, *Adding Authentication with JWT*, teaches you about JSON Web Tokens, a way to add authentication to web applications. It also provides instructions on how to set up multiple routes using React Router.

Chapter 7, *Improving the Load Time Using Server-Side Rendering*, covers benchmarking an application and teaches you about Web Vitals. Then, it gives instructions on how to implement a way to render React components on a server from scratch and how to pre-fetch data on the server.

Chapter 8, *Making Sure Customers Find You with Search Engine Optimization*, focuses on how to optimize an app to be found by search engines, such as Google or Bing. Additionally, it provides information on

how to create meta tags for easier integration with various social media sites.

Chapter 9, *Implementing End-to-End Tests Using Playwright*, introduces Playwright as a tool for writing end-to-end tests, which automatically performs actions in an app to find out whether your code still runs as designed after making changes. It also covers how to run Playwright in CI using GitHub Actions.

Chapter 10, *Aggregating and Visualizing Statistics Using MongoDB and Victory*, provides instructions on how to collect events in an app. Then, it teaches you how to aggregate data with MongoDB to generate summary statistics, such as the number of views or session duration. Finally, it covers creating graphs to visualize those aggregated statistics using the Victory library.

Chapter 11, *Building a Backend with a GraphQL API*, introduces GraphQL as an alternative to REST APIs, and you will learn when it is useful to use and how to implement it in a backend.

Chapter 12, *Interfacing with GraphQL on the Frontend Using Apollo Client*, teaches you how to use Apollo Client on the frontend to interface with the previously implemented GraphQL backend.

Chapter 13, *Building an Event-Based Backend Using Express and Socket.IO*, introduces an event-based architecture, which is useful for apps that deal with real-time data, such as collaborative applications (Google Docs or an online whiteboard) or financial applications (Kraken crypto exchange). It teaches you about WebSockets and how to use Socket.IO to implement an event-based backend.

Chapter 14, *Creating a Frontend to Consume and Send Events*, implements a frontend for the previously created event-based backend and interfaces with it using Socket.IO.

Chapter 15, *Adding Persistence to Socket.IO Using MongoDB*, teaches you how to properly integrate a database into an event-based app to persist (and later replay) events.

Chapter 16, *Getting Started with Next.js*, introduces Next.js as an enterprise-ready full-stack web application framework for React. It highlights the differences between using a framework and a simple bundler such as Vite. It also teaches you about the Next.js App Router, a new paradigm for defining routes and pages.

Chapter 17, *Introducing React Server Components*, teaches you about a new concept in React, Server Components, allowing you to directly integrate React apps with a database without needing a REST or GraphQL API. Additionally, it teaches you about Server Actions, which allow you to call functions on the server via the frontend.

Chapter 18, *Advanced Next.js Concepts and Optimizations*, dives deeper into the Next.js framework, providing information on how caching works in Next.js and how it can be used to optimize applications. It also teaches you about defining API routes in Next.js and how to add metadata for search engine optimization. Lastly, it teaches you how to optimally load images and fonts in Next.js.

Chapter 19, *Deploying a Next.js App*, teaches you two ways to deploy a Next.js app. The easiest way is using the Vercel platform, with which we can quickly get our app up and running. However, it also teaches you how to create a custom deployment setup using Docker.

Chapter 20, *Diving Deeper into Full-stack Development*, briefly covers various advanced topics that have not been covered in this book yet. It starts with an overview of other full-stack frameworks and then summarizes concepts such as maintaining large-scale projects, optimizing the bundle size, an overview of UI libraries, and advanced state management solutions.

To get the most out of this book

Software/hardware covered in the book

Operating system requirements

Node.js v20.10.0

Windows, macOS, or Linux

Git v2.43.0

Visual Studio Code v1.84.2

Docker v24.0.6

Docker Desktop v4.25.2

MongoDB Shell v2.1.0

If you are using the digital version of this book, we advise you to type the code yourself or access the code from the book's GitHub repository (a link is available in the next section). Doing so will help you avoid any potential errors related to the copying and pasting of code.

Download the example code files

You can download the example code files for this book from GitHub at <https://github.com/PacktPublishing/Modern-Full-Stack-React-Projects>. If there's an update to the code, it will be updated in the GitHub repository.

We also have other code bundles from our rich catalog of books and videos available at <https://github.com/PacktPublishing/>. Check them out!

Code in Action

The *Code in Action* videos for this book can be viewed at <https://packt.link/VINfo>.

Conventions used

There are a number of text conventions used throughout this book.

Code in text: Indicates code words in text, database table names, folder names, filenames, file extensions, pathnames, dummy URLs, user input, and Twitter handles. Here is an example: “First, you need to create a **robots.txt** file to allow search engines whether they are allowed to crawl parts of your website, and which parts they are allowed to crawl.”

A block of code is set as follows:

```
export const getPostById = async (postId) => {  
  const res = await fetch(`${import.meta.env.VITE_BACKEND_URL}/posts/${postId}`  
  return await res.json()  
}
```

When we wish to draw your attention to a particular part of a code block, the relevant lines or items are set in bold:

```
{fullPost ? (  
  <h3>{title}</h3>  
) : (  
  <Link to={` /posts/${_id}`}>  
    <h3>{title}</h3>  
  </Link>  
)}
```

Any command-line input or output is written as follows:

```
$ npm install node-emoji
```

Bold: Indicates a new term, an important word, or words that you see on screen. For instance, words in menus or dialog boxes appear in **bold**. Here is an example: “Connect to the database, then expand the **Playgrounds** section (if it is not expanded already) and click on the **Create New Playground** button.”

TIPS OR IMPORTANT NOTES

Appear like this.

Get in touch

Feedback from our readers is always welcome.

General feedback: If you have questions about any aspect of this book, email us at customercare@packtpub.com and mention the book title in the subject of your message.

Errata: Although we have taken every care to ensure the accuracy of our content, mistakes do happen. If you have found a mistake in this book, we would be grateful if you would report this to us. Please visit www.packtpub.com/support/errata and fill in the form.

Piracy: If you come across any illegal copies of our works in any form on the internet, we would be grateful if you would provide us with the location address or website name. Please contact us at copyright@packt.com with a link to the material.

If you are interested in becoming an author: If there is a topic that you have expertise in and you are interested in either writing or contributing to a book, please visit authors.packtpub.com.

Share Your Thoughts

Once you've read *Modern Full-Stack React Projects*, we'd love to hear your thoughts! Please [click here to go straight to the Amazon review page](#) for this book and share your feedback.

Your review is important to us and the tech community and will help us make sure we're delivering excellent quality content.

Download a free PDF copy of this book

Thanks for purchasing this book!

Do you like to read on the go but are unable to carry your print books everywhere?

Is your eBook purchase not compatible with the device of your choice?

Don't worry, now with every Packt book you get a DRM-free PDF version of that book at no cost.

Read anywhere, any place, on any device. Search, copy, and paste code from your favorite technical books directly into your application.

The perks don't stop there, you can get exclusive access to discounts, newsletters, and great free content in your inbox daily

Follow these simple steps to get the benefits:

1. Scan the QR code or visit the link below



<https://packt.link/free-ebook/978-1-83763-795-9>

2. Submit your proof of purchase
3. That's it! We'll send your free PDF and other benefits to your email directly